ASSIGNMENT 5

Database Creation and Table Implementation

HOSPITAL MANAGEMENT SYSTEM

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Course: BCA (Hons) | Subject: Database Management System

1. Objective

Create the hospital_management database and implement the eight tables in the finalized relational schema. Run the statements in order in MySQL 8. The primary and foreign keys are defined in the CREATE TABLE statements.

2. Database and table creation SQL

```
CREATE DATABASE IF NOT EXISTS hospital_management;
USE hospital_management;

CREATE TABLE Departments (   department_id INT PRIMARY KEY,
department_name VARCHAR(80) NOT NULL UNIQUE,   location
VARCHAR(100) NOT NULL ); CREATE TABLE Patients (   patient_id
INT PRIMARY KEY,   name VARCHAR(100) NOT NULL,   gender
VARCHAR(10) NOT NULL,   dob DATE NOT NULL,   phone
VARCHAR(15) NOT NULL,   address VARCHAR(255) NOT NULL );
CREATE TABLE Doctors (   doctor_id INT PRIMARY KEY,
department_id INT NOT NULL,   name VARCHAR(100) NOT NULL,
specialization VARCHAR(80) NOT NULL,   phone VARCHAR(15) NOT
NULL,   email VARCHAR(120) NOT NULL UNIQUE,   CONSTRAINT
fk_doctor_department FOREIGN KEY (department_id)
    REFERENCES Departments(department_id)
); CREATE TABLE Appointments (
appointment_id INT PRIMARY KEY,
patient_id INT NOT NULL,
doctor_id INT NOT NULL,
appointment_date DATE NOT NULL,
appointment_time TIME NOT NULL,
  status VARCHAR(30) NOT NULL DEFAULT 'Scheduled',   CONSTRAINT chk_appointment_status CHECK (status IN
('Scheduled','Completed','Cancelled','No-show')),
  CONSTRAINT fk_appointment_patient FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),
  CONSTRAINT fk_appointment_doctor FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)
);
```



```
CREATE TABLE Medicines (   medicine_id INT PRIMARY
KEY,   medicine_name VARCHAR(120) NOT NULL,
manufacturer VARCHAR(100) NOT NULL,   price
DECIMAL(10,2) NOT NULL,   CONSTRAINT
chk_medicine_price CHECK (price >= 0)
); CREATE TABLE Prescriptions (   prescription_id INT PRIMARY
KEY,   appointment_id INT NOT NULL,   medicine_id INT NOT
NULL,   dosage VARCHAR(50) NOT NULL,   duration VARCHAR(50)
NOT NULL,   quantity INT NOT NULL,   CONSTRAINT
chk_prescription_quantity CHECK (quantity > 0),
  CONSTRAINT uq_appointment_medicine UNIQUE (appointment_id, medicine_id),
  CONSTRAINT fk_prescription_appointment FOREIGN KEY (appointment_id) REFERENCES Appointments(appointment_id),
  CONSTRAINT fk_prescription_medicine FOREIGN KEY (medicine_id) REFERENCES Medicines(medicine_id)
); CREATE TABLE Bills (   bill_id INT PRIMARY KEY,
appointment_id INT NOT NULL,   bill_date DATETIME NOT
NULL DEFAULT CURRENT_TIMESTAMP,   total_amount
DECIMAL(10,2) NOT NULL,
  payment_status VARCHAR(30) NOT NULL DEFAULT 'Pending',
CONSTRAINT chk_bill_amount CHECK (total_amount >= 0),
  CONSTRAINT chk_payment_status CHECK (payment_status IN ('Pending','Paid','Partially Paid','Refunded')),
CONSTRAINT fk_bill_appointment FOREIGN KEY (appointment_id) REFERENCES Appointments(appointment_id)
```

```
); CREATE TABLE Feedbacks (   feedback_id INT PRIMARY KEY,
patient_id INT NOT NULL,   doctor_id INT NOT NULL,   rating INT
NOT NULL,   comments VARCHAR(500),   CONSTRAINT
chk_feedback_rating CHECK (rating BETWEEN 1 AND 5),
  CONSTRAINT fk_feedback_patient FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),
  CONSTRAINT fk_feedback_doctor FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)
);
```

3. Verify that all tables were created

```
USE hospital_management;
SHOW TABLES;
DESCRIBE Patients;
DESCRIBE Appointments;
DESCRIBE Prescriptions;
DESCRIBE Bills;
```

Expected verification

SHOW TABLES should list Appointments, Bills, Departments, Doctors, Feedbacks, Medicines, Patients and Prescriptions. DESCRIBE displays each column, data type, nullability, key and default value. Exact output formatting depends on the MySQL client.

4. Implementation notes

- Create parent tables before child tables so foreign-key references exist.
- Use InnoDB to support foreign keys and transactions; it is the default engine in modern MySQL installations.
- The CHECK constraints are enforced in MySQL 8.0.16 and later.
- For a fresh run, use an empty database. If rerunning, drop dependent tables in reverse dependency order or create aseparate test database.

Conclusion

The DDL creates the eight-table relational structure and establishes the relationships specified in Assignment 2. Assignment 6 adds focused explanations and verification of integrity constraints.

ASSIGNMENT 5 — OUTPUTS AND VERIFICATION

Output 1 — SHOW TABLES

Tables_in_hospital_management
Appointments
Bills
Departments
Doctors
Feedbacks
Medicines
Patients
Prescriptions

Output 2 — Key column checks from DESCRIBE

Table	Column	Type	Key	Null / Default (summary)
Patients	patient_id	INT	PRI	NO
Doctors	doctor_id	INT	PRI	NO
Doctors	department_id	INT	MUL / FK	NO
Appointments	patient_id	INT	MUL / FK	NO
Appointments	doctor_id	INT	MUL / FK	NO
Appointments	status	VARCHAR(30)		NO / default Scheduled
Prescriptions	appointment_id	INT	MUL / FK	NO
Prescriptions	medicine_id	INT	MUL / FK	NO
Bills	appointment_id	INT	MUL / FK	NO
Feedbacks	rating	INT		NO

Result

Expected result: eight tables are listed. DESCRIBE confirms primary keys, foreign-key columns and defaults. Run SHOW CREATE TABLE for the full constraint definitions.